Superblocks: case study overview

Description: The Superblocks model in Barcelona is an urban environment regeneration intervention targeting adult population aged 30 ~~20~~ years and above. The model introduces functional and structural changes to the streets and the city layout in order to reshape modern cities to be more people centric and less reliant on motor vehicles. Superblocks aims to achieve several objectives including a reduction in environmental pollution (air and noise pollution) as well as boost levels of physical activity.

Best practice assessment:

OECD Framework assessment of Superblocks

Criteria	Assessment
Effectiveness	Over 16 545 life years and 25 482 disability-adjusted life years (DALYs) gained by 2050 if expanded across the whole of Spain By increasing total physical activity in city dwellers, Superblocks can prevent several forms of chronic diseases, on mental health, followed by cardiovascular diseases, diabetes and finally cancers related physical inactivity
Efficiency	Superblocks intervention is found to be cost-effective across all of OECD and EU27 countries at the most conservative assumed cost per capita per year, however at the least conservative cost per capita per year the intervention is cost effective across 40% of OECD and EU27 countries.
Equity	Superblocks aims to promote healthier way of life and reduce inequity in the low socioeconomic status neighbourhoods (SES) of Barcelona through urban design that facilitates access to safe public spaces, cleaner environment and social cohesion.
Evidence-base	The quality of evidence used for this case study is "moderate" in areas related to data collection methods, and selection bias. Similar to most public health studies, the design of the intervention does not allow blinding and measurement
Extent of coverage	Superblocks reaches approximately 8% of adults aged at least 20 years in Spain.

Enhancement options: to *enhance the effectiveness*, policy makers incorporate at the national level changes to the cycling policies, parking and road pricing regulation in order to prioritise the use of bikes and other non-motorised methods of transport. To *enhance the evidence-base*, health and urban environment policymakers makers could spearhead the use of more advanced measuring tools for changes in city dwellers physical activity levels in order to capture shifts in daily physical activity. To *enhance equity*, administrators could implement lessons learned from previous implementation and get ahead of equity considerations that became known. To *enhance extent of coverage* several strategies are available for the administrators, such as active involvement of stakeholders in planning and execution.

Transferability: the Superblocks model targets risk factors prevalent in all OECD countries, therefore the intervention is likely to receive strong political support, which is necessary when considering which interventions to transfer. Further, similar urban models operate or are planned in several European cities such as Paris, London and Hamburg indicating they are transferable when adapted to the local context.

Conclusion: the Superblocks approach has the potential to significantly reduce the incidence of non-communicable disease when scaled-up across Spain and transferred to other OECD and non-OECD European countries.

1. Case study: Barcelona Superblocks

1.1. Intervention description

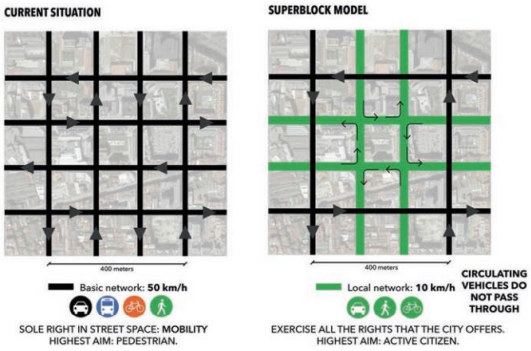
Major cities across the world have seen a marked increase in the use of cars over the past century with rapid expansions of industrialisation and urbanisation, including Barcelona. As of 2020, nearly four in ten people in Barcelona used a private car for journeys, which is roughly the same as those who walk. The high number of people who walk is partly attributed to the city's climate, however, it is interesting to note that rates of cycling are low with just 2% of people utilising this form of transport (Deloitte, 2020[1]).

Existing architecture and urban planning in many European cities was not built to accommodate such a large number of vehicles – for example, in regards to road space, parking, speed regulations and other safety measures. Consequently, the construction of additional roads to allow more motorised vehicles has limited the availability of urban green spaces (e.g. parks, natural environments, open spaces and bike lanes). The loss of urban green spaces reduces opportunities to engage in physical activity and contributes to high levels of environmental pollution, which includes air pollution, noise and anthropogenic heat (Nieuwenhuijsen and Khreis, 2018[2]).

To address health and environmental concerns caused by the rise in the number of private and commercial vehicles in Barcelona, in 2005, the Urban Ecology Agency (BCNEcologia) within the Barcelona City Council introduced the Superblocks initiative. Superblocks aims to change the organisational fabric of the city by adopting an urban environment that promotes mobility, public space, biodiversity and social cohesion for a healthier city. Through this transformation, the city prioritises space for residents over the space for cars (Energy Cities, 2016[3]).

The concept of Superblocks builds off Barcelona's existing grid-like urban design by changing the way traffic circulation is organised (Riba, 2021[4]). Specifically, the initiative breaks Barcelona into a number of superblocks comprising 9 ~~4-5~~ blocks of enclosed inner streets that permit cars of residents to come in and out at a maximum speed of 10km/hr, while restricting outside traffic from passing through the superblock (with the of emergency vehicles) (Figure 1.1).

Figure 1.1. Road hierarchy in the new Superblock model



Source: Rueda, S. (2018) Image taken from OECD (2021[5]), "Transport Strategies for Net-Zero Systems by Design", https://dx.doi.org/10.1787/0a20f779-en.

This new urban design has several objectives such as improving urban mobility, preserving biodiversity, encouraging social cohesion and collaboration, as well as reducing noise, air pollution and heat as summarised in Table 1.1. Reducing heat, in particular, is of growing interest to policymakers given the rise of the "urban heat island effect" (see Box 1.1).

Table 1.1. The objectives of Barcelona's Superblocks initiative

Objective	Description
Improve sustainable urban mobility	Establish new transportation networks that reduce inter-city traffic circulation intended to relieve car congestion in residential areas, decrease the attractiveness of car use, and encourage both pedestrian and bicycle mode of transportations. Additionally establish the predominance of public transportation through improved services by providing better connectivity, faster lines, as well as increasing frequency and safety.
Preservation of biodiversity within the city	Improve the conditions for existing biodiversity by reducing the number of vehicles. In addition, promote green spaces through green alleys, gardens, and parks that can contribute to the restoration of ecosystems
Promotion of sustainable energy production and consumption	Reduce energy consumption through improvements in efficient use of energy (buildings, public lighting), promote the use and production of renewable energy sources and selective waste collection, as well as reducing the use of drinking water for non-drinking purposes (e.g. cleaning streets).
Reduction of noise, air pollution, carbon footprint and urban heat	Reduce the level of noise, greenhouse gas emissions and other air pollutants. By doing so, Superblocks discourages heat accumulation (also referred to as the "urban heat island effect" (see Box 1.1).
Revitalisation of streets and public spaces	Foster common leisure spaces, street vitality and create space for the development of small local enterprises
Encouragement of social cohesion and collaboration	Encourage cohesion and collaboration by creating new job opportunities (via small local businesses), new partnerships between the inhabitants and bigger structures and organisations, providing necessary space for local meetings, and educational workshops.
Introduction of participation and new models of governance	Promotion and creation of opportunities for city dwellers to participate and lend their voices to the decisions regarding their neighbourhoods while further encouraging collaborations between the locals and diverse organisations that facilitate desired changes.

Source: Rueda, S. (2007[6]), "Barcelona, compact and complex Mediterranean city: A more sustainable vision for the future", http://bcnecologia.net/en/publications/barcelona-compact-and-complex-mediterranean-city-more-sustainable-vision-future; Superblocks (2022[7]), "Welcome to Superilles | Superilles", https://ajuntament.barcelona.cat/superilles/en/.

Box 1.1. Urban heat island effect

To accommodate a large number of residents, today's modern urban environments have replaced natural land with pavements, buildings and heat absorbent surfaces. This has led to a rise in the number of urban heat islands (UHIs), which refer to urban or metropolitan areas that are significantly warmer than surrounding rural areas (Santamouris, 2001[8]; Grimmond, 2007[9]; Oke et al., 2017[10]).

UHIs pose a significant health risk to the population by increasing the number of heat-related deaths and heat-related illnesses, such as troubles breathing, exhaustion, cramps and strokes. The negative effects of UHI disproportionality impact vulnerable groups such as the elderly, young children, as well as those on low incomes (e.g. due to poor housing with no air conditioning). During the COVID-19 pandemic, UHIs also posed a danger for health professionals who are required to extensive personal protective equipment (Bose-O'Reilly et al., 2021[11]).

Effectiveness

The effectiveness of Barcelona's Superblocks initiative is based on modelling estimates produced by OECD's Strategic Planning for Public Health NCD (SPHeP-NCD) model. The analysis focuses on Superblocks' impact on changes in physical activity due to a change in transport preferences (e.g. from private car use to walking). Results from this analysis underestimate Superblocks' total impact, given it does not take into account changes in health outcomes due to improvements in environmental pollution (i.e. air and noise pollution, and anthropogenic heat).

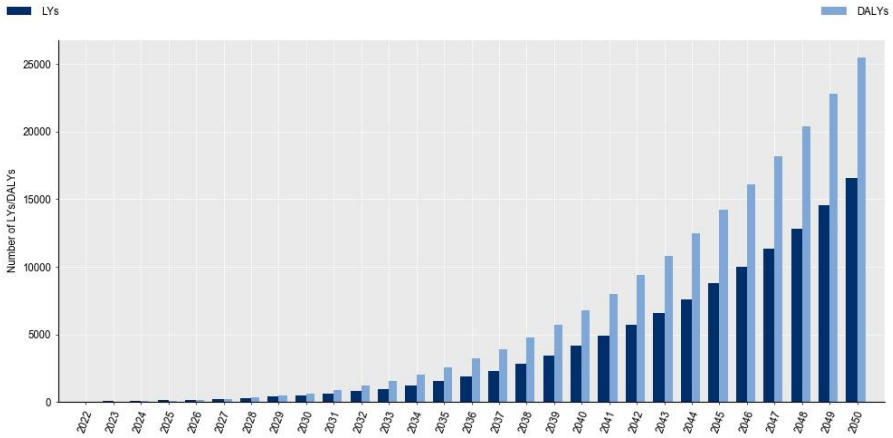
Results are available at the national level for Spain and all other OECD and EU countries over years 2022-2050. That is, the model assumes Superblocks has been scaled up to eligible populations across the country. Further details on model inputs and assumptions are available in Annex A.

In addition to reporting results from the modelling exercise, this section includes a summary of Superblocks' real-world impact on environmental pollution.

Spain

The model estimates that 16 545 life years (LYs) would be gained between 2022-50 by scaling-up Superblocks to all eligible populations in Spain. The number of disability-adjusted life years (DALYs) gained is even higher at 25 482 over the same period (Figure 1.2).

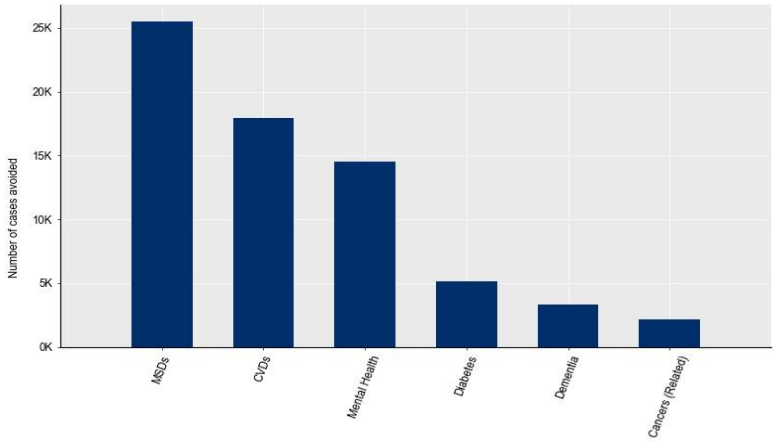
Figure 1.2. Cumulative number of LYs and DALYs gained (2022-50) – Spain



Source: OECD analysis based on the OECD SPHeP-NCD model, 2022.

In gross terms, Superblocks' would have the largest impact on musculoskeletal disorders with 25 537 cases and 14 504 mental ill-health cases avoided between now and 2050 (Figure 1.3). Other diseases affected by Superblocks' include CVDs (17 974 cases avoided), diabetes (5 164 cases) and several related cancers (2 153 cases).

Figure 1.3. Cumulative number of disease cases avoided by 2050 – Spain

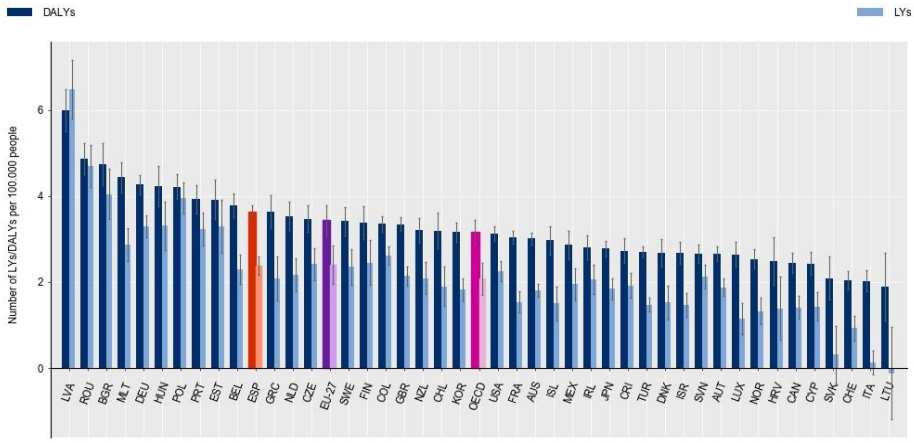


Note: Related cancers include colorectum, breast, esophageal and liver cancer. The black lines represent 95% confidence intervals.
Source: OECD analyses based on the OECD SPHeP-NCDs model, 2022.

OECD and non-OECD European countries

Transferring Superblocks to all OECD and EU27 countries is estimated to result in 2.4 and 18.4 LYs gained per 100 000 people, respectively, on average per year between 2021-50 (ranging from 6.5 in Latvia to 0.14 in Italy). For DALYs, gains are even higher at 3.2 for OECD and 3.4 for EU27 countries (Figure 1.4).

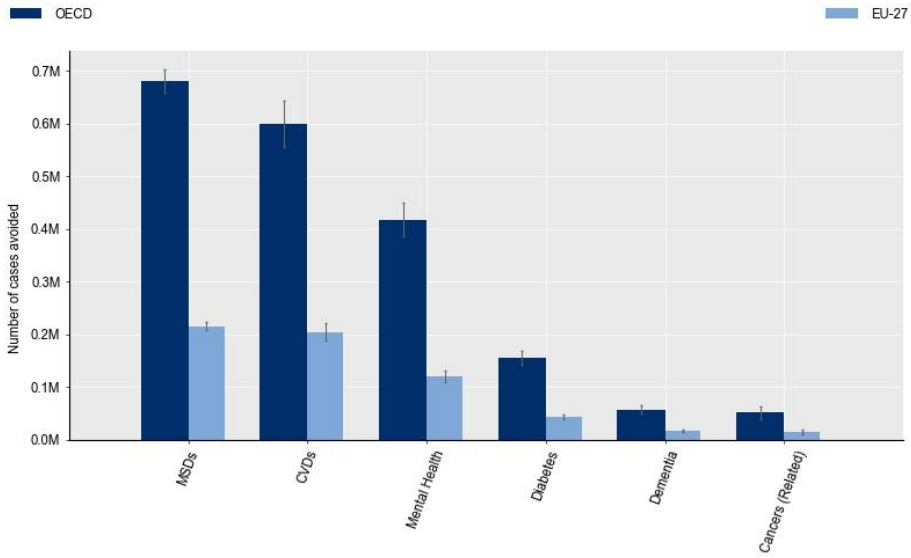
Figure 1.4. LYs and DALYs gained annually per 100 000 people, 2021-50 – all countries



Note: The values for Lithuania are not significant. The black lines represent 95% confidence intervals.
Source: OECD analyses based on the OECD SPHeP-NCDs model, 2022.

Overall, Superblocks would have the greatest impact on MSDs with the intervention estimated to reduce the number of cases by 680 589 and 214 674 among OECD and EU27 countries, respectively, between 2022 and 2050 (Figure 1.5). Across all countries, Superblocks would also reduce the number of CVD cases by 599 404 cases, mental health disorders cases by 417 583, diabetes cases by 154 568, and cases of cancer related to physical inactivity by 51 143.

Figure 1.5. Total disease cases avoided, between 2021 and 2050 – OECD and EU27 countries



Note: The grey lines represent 95% confidence intervals.
Source: OECD analyses based on the OECD SPHeP-NCDs model, 2022

Real-world outcomes

Real-world effects observed by previous evaluations of Superblocks complement findings from the modelling exercise. Namely the impact of Superblocks on noise and air pollution.

Noise pollution

Noise pollution is a widespread public health concern, especially among city dwellers. It refers to the accumulation of noise (often generated from cars) that negatively affect the health and wellbeing of urban inhabitants.

Long-term exposure to environmental noise in cities can result in several negative side effects ranging from sleep disturbance to poorer cardiovascular and metabolic health as well as cognitive impairment in children (European Environment Agency, 2020[12]). According to findings in recent European Environment Agency, 20% of Europeans are exposed to the long-term noise pollution that is harmful to their health, which corresponds to 100 million people (European Environment Agency, 2020[12]). In Barcelona, the figure is even higher with 50% of inhabitants living with noise levels above the safe levels (53dBs) as indicated by the WHO (Ajuntament de Barcelona, 2017[13]; Berglund, Lindvall and Schwela, 1999[14]).

An evaluation of the superblock in San Antoni examined its impact on noise pollution using a pre/post study design. Between 2017 and 2019, the authors found the average decibel level during the day fell by 3.5 points, which equates to a fall of 5.2%. However, it remainder the same during the night (Agència de Salut Pública de Barcelona, 2021[15]).

Air pollution

Air quality is a strong determinant of health outcomes. Air pollution comes from various sources in an urban setting such as traffic combustion emissions, domestic fuel burning, agriculture, natural dust and salt, and industrial activity (Karagulian et al., 2015[16]).

The dangers of air pollution to human health comes from several contaminants: particulate matter ($PM_{2.5}$ and PM_{10} – defined as aerodynamic diameter of the PM), ground level of ozone (O_3), carbon monoxide (CO), sulphur dioxide (SO_2), and nitrogen dioxide (NO_2) (Brook et al., 2004[17]; Sheehan et al., 2016[18]). The chronic exposure to these elements, as is the case of city inhabitants, increases both morbidity and mortality, for example, from asthma, COPD, stroke, lung cancer, impairment to cognitive function, along with other respiratory conditions (Dominski et al., 2021[19]). According to WHO estimates, 7 million deaths occur every year due to the pollution of outdoor weather, with 90% of all deaths among people living in areas that exceed the WHO guideline limits on air quality (World Health Organization, 2018[20]).

Air pollution is also an economic concern with costs around the world estimated at 4.6 trillion USD per year, which represents 6.2% of the world economic output through both productivity loss as well as welfare costs (Landrigan et al., 2018[21]).

Superblocks aims to reduce air pollution by decreasing urban traffic. The impact of superblocks on air pollution as recorded in San Atoni and Horta are summarised in the table below. In summary, the level of air pollution fell among all indicators with decreases ranging from -7% to -93%.

Table 1.2. Average annual air pollution levels in two Superblocks pre and post implementation

Air Pollution (annual average) µg/m³	Before Superblocks (2017-2018)	After Superblocks (2018-2019)	Percentage reduction	Limits (annual mean)
NO_2 *San Antoni*	57	23	-60%	WHO limit:10 µg/m³ EU limits :40 µg/m³
PM_{10} *San Antoni*	25.6	23	-10%	WHO limit : 15 µg/m³ EU limit: 40 µg/m³
NO_2 *Horta*	35.6	24.6	-93%	WHO limit:10 µg/m³ -EU limits :40 µg/m³
PM_{10} *Horta*	19.7	14.7	-25%	WHO limit : 15 µg/m³ EU limit: 40 µg/m³
$PM_{2.5}$ *Horta*	12.1	11.2	-7%	WHO limit : 5 µg/m³ EU limit: 20 µg/m³

Note: The measurements were averaged out throughout the year from before and after implementation of Superblocks.
Source: WHO (2021[22]), "Ambient (outdoor) air pollution", https://www.who.int/news-room/fact-sheets/detail/ambient-(outdoor)-air-quality-and-health; European Commission (2020[23]), "Standards - Air Quality - Environment", https://ec.europa.eu/environment/air/quality/standards.htm; Agència de Salut Pública de Barcelona (2021[15]), "Informe de resultats dels efectes mediambientals i en la salut del model Superilles a Barcelona. Projecte Salut Als Carrers (SAC).", https://www.aspb.cat/.

1.1. OECD Best Practices Framework assessment

This section analyses the Superblocks against the five criteria within OECD's Best Practice Identification Framework – Effectiveness, Efficiency, Equity, Evidence-base and Extent of coverage (see the box below for a high-level assessment of Superblocks). Further details on the OECD Framework can be found the Best Practice case study guide.

Box 1.2. Best practice assessment overview: Superblock design, Barcelona

Effectiveness

- Superblocks model has shown to increase the levels of overall physical activity in city dwellers aged 20 years and above
- According to OECD simulations, Superblocks would lead to 16 545 life years and 25 482 disability-adjusted life years (DALYs) gained by 2050 if expanded across the whole of Spain
- Across all OECD and EU27 countries, Superblocks would have the largest gross impact on mental health, followed by cardiovascular diseases, diabetes and finally cancers related physical inactivity

Efficiency

- When expanded across the whole of Spain it is estimated Superblocks will accumulate health expenditure savings of EUR 4.04 per person by 2050
- When transferred to all OECD and EU27 countries, savings equivalent to 0.01% of total health expenditure per year are expected
- For the majority of OECD and EU27 countries, Superblocks is not only cost effective, but also cost saving

Equity

- Superblocks has a process in place to adapt environmental changes to suit the needs of priority populations – e.g. cyclists, pedestrians and low-emissions vehicles, especially for those with a low socioeconomic status

Evidence base

- A recent study evaluated the impact of Superblocks supports the hypothesis that Superblocks increases total physical activity among city inhabitants and reduces air and noise pollution
- The study used modelling approach to evaluate long-term the impact of Superblocks using real-life collected and surveyed data

Extent of coverage

- Superblocks reaches approximately 8% of adults aged at least 20 years of age in Spain.

Comentado [CJE1]: This fits better under effectiveness as it discusses other benefits of Superblocks to date. In the final version, this will need to be cut down quite a lot to just the key messages as the modelling will be the focus.

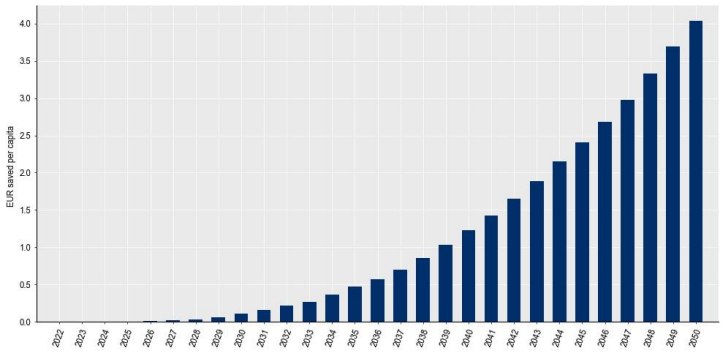
Efficiency

The analysis from OECD's microsimulation model details efficiency gains from Superblocks due to an improvement in overall health – i.e. reduced healthcare costs and greater labour productivity. Similar to effectiveness estimates, results are presented at the national level for Spain and all other OECD and non-OECD European countries.

Spain

By increasing physical activity levels, Superblocks can reduce healthcare related costs. Over the modelled period of 2022-50, the OECD-SPHeP NCD model estimates that Superblocks would lead to cumulative health expenditure savings of EUR 4.04 per person (Figure 1.6) or EUR 0.26 per person, per year. Cost savings however are to an extent offset by intervention operating costs (see Table 1.3).

Figure 1.6. Cumulative health expenditure savings per person, EUR, 2021-2050 – Spain

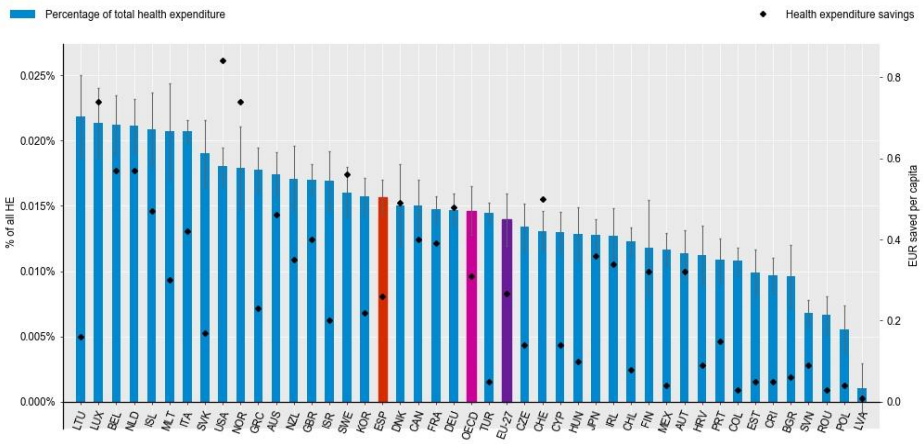


Source: OECD analyses based on the OECD SPHeP-NCDs model, 2022.

OECD and non-OECD European countries

Average annual health expenditure (HE) savings as a proportion of total HE is 0.01% for both OECD and EU27 countries (Figure 1.7). On a per capita basis, this translates into average annual savings of EUR 0.31 and EUR 0.27 for OECD and EU27 countries, respectively.

Figure 1.7. Health expenditure (HE) savings as a % of total HE and per capita (EUR), average 2021-50 – all countries



Note: The black lines represent 95% confidence intervals.
Source: OECD analyses based on the OECD SPHeP-NCDs model, 2022.

The table below provides information on intervention costs, total health expenditure savings and the cost per DALY gained in local currency for OECD and non-OECD European countries. Superblocks intervention

is found to be cost-effective across all of OECD and EU27 countries when using the most conservative assumed cost per capita per year, however at the least conservative cost per capita per year the intervention is cost effective across 40% of OECD and EU27 countries.

Comentado [CJE2]: I don't think this is true.

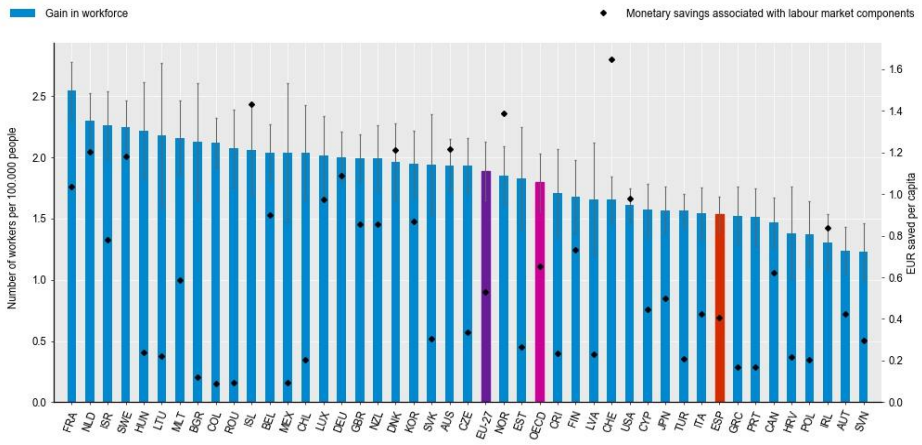
Table 1.3. Cost effectiveness figures in local currency – all countries

Country	Local currency	Intervention costs per capita, average per year	Total health expenditure savings, 2022-50	Cost per DALY gained*
Australia	AUD	1.46	21766310	Cost effective
Austria	EUR	0.76	2864649	Cost effective
Belgium	EUR	0.75	6935498	Cost effective
Bulgaria	BGN	0.69	663469	Cost effective
Canada	CAD	1.20	24408951	Cost effective
Chile	CLF	1.15	5233958	Cost effective
Colombia	COP	395.08	1318578878	Cost effective
Costa Rica	CRC	1336.41	6505758311	Cost effective
Croatia	HRK	338.15	183253681	Cost effective
Cyprus	EUR	0.60	181031	Cost effective
Czech Republic	CZK	12.41	36612710	Cost effective
Denmark	DKK	0.74	37529579	Cost effective
Estonia	EUR	6.59	22154000	Cost effective
Finland	EUR	0.62	11897994	Cost effective
France	EUR	0.54	62429	Cost effective
Germany	EUR	0.84	1816639	Cost effective
Greece	EUR	0.72	26769234	Cost effective
Hungary	HUF	0.68	24984208	Cost effective
Iceland	ISK	0.55	2361342	Cost effective
Ireland	EUR	3.24	2490239	Cost effective
Israel	ILS	141.85	297542937	Cost effective
Italy	EUR	0.80	1784101	Cost effective
Japan	JPY	139.21	24011039	Cost effective
Korea	KRW	3.67	8824010	Cost effective
Latvia	EUR	0.66	24414038	Cost effective
Lithuania	EUR	102.63	4970490673	Cost effective
Luxembourg	EUR	860.20	14569002834	Cost effective
Malta	EUR	0.45	426013	Cost effective
Mexico	MXN	0.84	518364	Cost effective
Netherlands	EUR	0.49	16490	Cost effective
New Zealand	NZD	9.28	132237849	Cost effective
Norway	NOK	0.57	129174	Cost effective
Poland	PLN	0.78	10025066	Cost effective
Portugal	EUR	9.67	44578716	Cost effective
Romania	RON	1.43	3135682	Cost effective
Slovakia	EUR	1.75	5821189	Cost effective
Slovenia	EUR	0.56	1489388	Cost effective
Spain	EUR	1.67	2420752	Cost effective
Sweden	SEK	0.52	886820	Cost effective
Switzerland	CHE	0.56	168538	Cost effective
Turkey	TRY	8.79	65057222	Cost effective
United Kingdom	GBP	1.88	28957457	Cost effective
United States	USD	0.99	347426010	Cost effective

Note: *Cost per DALY gained is measured using total intervention costs less total health expenditure savings divided by total DALYs gained over the period 2022-50. Costs and benefits have been discounted at a rate of 3%. This table shows the cost results for the most conservative scenario.
Source: OECD analyses based on the OECD SPHeP-NCDs model, 2022.

The reduction in chronic diseases resulting from Superblocks has, in turn, an impact on labour market participation and productivity. By reducing chronic disease incidence, Superblocks is expected to lead to increases in employment and reductions in absenteeism, presenteeism, and early retirement. Converting these labour market outputs into full-time equivalent (FTE) workers, it is estimated that OECD and EU27 countries will gain 1.89 FTE and 1.79 FTE per 100 000, respectively, for working age people per year between 2022 and 2050. In monetary terms, this translates into average per capita labour market savings of EUR 0.65 for OECD and EUR 0.53 for EU27 countries (Figure 1.8).

Figure 1.8. Labour market workforce gains and savings, average per year, 2021-50 – all countries



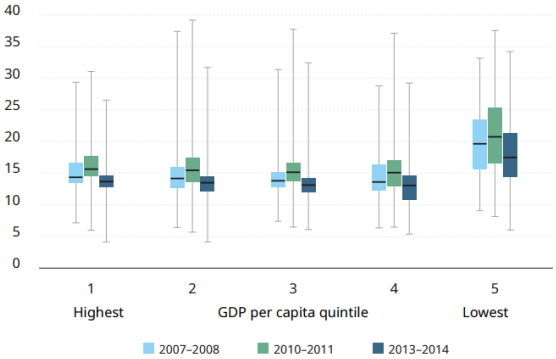
Source: OECD analyses based on the OECD SPHeP-NCDs model, 2021.

Equity

At a high-level, Superblocks aims to promote health equity by providing everyone, regardless of income, with access to public spaces, parks and other green areas, and a community of neighbours to improve social interaction. Each of the aforementioned benefits is associated with better physical and mental health thereby reducing existing health inequalities.

Superblocks has the potential to disproportionately benefit vulnerable groups, such as those with a low socioeconomic status, given these groups are more likely to live in highly polluted, noisy areas (e.g. more likely to live close to roads with heavier traffic as well as contaminated sites such as landfills) (see Figure 1.9). In addition, disadvantaged populations are more likely to suffer from underlying health problems, that are exacerbated by air pollution (e.g. CVDs) (WHO Europe, 2021[24]). Despite these benefits, policy makers should be wary of the potential negative impact rising from gentrification in the city (Palència et al., 2020[25]).

Figure 1.9. Population-weighted $PM_{2.5}$ concentration (µg/m3)



Source: Image taken from WHO Regional Office for Europe (2019[26]), "Environmental health inequalities in Europe. Second assessment report", https://www.euro.who.int/en/publications/abstracts/environmental-health-inequalities-in-europe.-second-assessment-report-2019.

Gentrification is a social phenomenon that results in the displacement of lower socioeconomic groups due to improvements to the urban environment (e.g. infrastructural attractiveness, communal spaces, access to greenspaces and reduced noise levels). A side effect of gentrification is a rise in property prices resulting from increased demand from local residents as well as domestic and foreign investors (e.g. converting living spaces into hotels or an Airbnb). Consequently, long-term inhabitants of the city with a lower socioeconomic status (SES) are pushed out of the city (Roberts, 2019[27]).

Gentrification can further affect low SES populations given work and education opportunities tend to be concentrated in the city. In addition, gentrification may widen existing health inequalities as it is associated with higher levels of mental ill-health due to severed social networks, instability, a lost sense of security and a feeling of helplessness (Mehdipanah et al., 2017[28]).

Evidence base

The section on "Evidence base" examines the quality of studies used to evaluate the impact of Superblocks to date, in particular regarding its effectiveness and efficiency. The inputs used to model the long-term health and economic impact of Superblocks are based on assumptions on how it may change individual level behaviour regarding active modes of transport. Therefore, it is important to interpret findings from this case study with caution, as inputs do not reflect findings from real-world data.

Assumptions used in the model are summarised in Annex A, and were drawn from Mueller et al's (2020[29]) study. A summary of inputs derived from this study are in the box below.

Box 1.3. Changing the urban design of cities for health: The superblock model – study by Mueller and colleagues, 2020

Mueller et al's study used quantitative tools along both real life and projected data in order to establish estimates on the shifts from motorised vehicle transport to more active and public modes of transportation due to Superblocks.

It was estimated that the share of motorised transport fell by 19.2% based on evaluations conducted in Urban Mobility Plan by Barcelona City Council and BCNecologia (Barcelona City Council, 2014[30]). The estimates for baseline physical activity in the population were obtained from Health Survey 2015/16 conducted among city residents, aged 20 years or older, while transport patterns (trips per day, distances, and time spent on mode of transport) were supported by transport data from Metropolitan Area of Barcelona (Direcció de Serveis de Mobilitat, 2016[31]).

The unit of measurement of METs (metabolic equivalent of task) was used to quantify projected increase in physical activity among city dwellers who would change from personal vehicle transport to either active or public transport. The measurements and estimates permitted to project the amounts of transport-related physical activity gained by the mode of transport: public services, cycling or walking.

Source: Mueller et al. (2020[29]), "Changing the urban design of cities for health: The superblock model", https://doi.org/10.1016/j.envint.2019.105132.

Extent of coverage

The city of Barcelona has, to date, implemented six superblocks covering 143 hectares of space and 33 000 residents (all of whom have equal access the superblocks' benefits).[1] A further three superblocks are being planned (OECD, 2021[5]).

There is significant scope to extend the coverage of this intervention, as there is space for a total of 503 superblocks in Barcelona.

1.2. Policy options to enhance performance

This section summarises policy options to enhance the performance of the Superblock approach in areas where the intervention currently operates. The policy options are also useful for policy makers in the process of, or interested in, implementing Superblocks (or a similar concept).

Enhancing effectiveness

Superblocks aims to increase active modes of transport by improving infrastructure and limiting the use of cars and other motor vehicles. Policy makers can further encourage residents to use active modes of transport by implementing complementary policies. Complementary policies relate to cycling, parking and road pricing schemes.

[1] The average number of residents within a superblock is between 5 000 and 6 000 (OECD, 2021[5]).

Cycling specific policies

To boost rates of cycling, policy makers could consider recommendations developed as part of the Pan-European's Plan (PEP) for Cycling Promotion Toolbox (The PEP Partnership on Cycling, 2021[32]). The Toolbox outlines several policies that encourage everyday bicycle use such as creating cycling working groups and a national cycling officer, setting up national funding schemes to promote cycling, introducing fiscal incentives for cycling, promotional and education campaigns, as well as integrating health- and cycling-related issues into educational activities. For further details, see Box 1.4, however, note the box does not cover all recommendations in the Toolbox.

Box 1.4. The PEP Partnership on Cycling: Toolbox of action to promote cycling

This box outlines a selection of recommendations within PEP's Tool of Action for Cycling Promotion. The list is not exhaustive given many of the recommendations, in particular those related to "including cycling in the planning process", have already been implemented in Barcelona, Spain.

Create strong cycling working groups and appoint a national cycling officer (NCO) (Recommendation: 1.2)

The NCO would play a key role in ensuring regular contact and exchange of ideas between different levels of stakeholders (e.g. health, transport, economic sectors) to ensure a better understanding of cycling needs and requirements. An NCO is of particular important if Superblocks is expanded to other parts of the country. In *Luxembourg*, the NCO is appointed by the Ministry of Mobility and Public Works and is required to review each infrastructure project to ensure the needs of cyclists and pedestrians have been taken into account.

Set up a national funding scheme to promote cycling (Recommendation 4.1)

A national funding scheme is important for supporting local and regional authorities promote cycling. In addition to public sector funds, a national funding scheme could explore opportunities for ongoing streams of funding from the private sector and other financial transport regulators (e.g. parking fees). The *United Nations Environment Programs currently recommends 20% of transport budgets be dedicated to walking and cycling*.

Introduce fiscal incentives for cycling (Recommendation 6.1)

One such example is offering residents a tax-free kilometric reimbursement for cycling to work. For example, *Belgium's Cycling Kilometric Allowance Scheme* (CKAS) is an intervention designed to promote active transport by providing financial incentives to employees who choose to cycle to work with their own bike. Employers who sign up to the scheme pay participating employees EUR 0.24 per kilometre cycled, which is tax-exempt.

Integrate health- and cycling-related issues into formal and informal education and awareness-raising activities (Recommendation 7.2)

Ministry of Health and Education should actively promote the health benefits associated with cycling as well as safety rules into the school curricula. For example, in *Denmark, the Netherlands and Germany*, children receive extensive training in safety and effective cycling techniques as part of their regular school curriculum. In fact, most children obtain the necessary knowledge and training by fourth grade (aged around 9-10 years old).

Incorporate cycling into road safety policies (Recommendation 7.3)

The needs of cyclists should be incorporated into road safety policies – covering topics such as infrastructure and speed managements as well as improving road users' behaviour through education, awareness raising and enforcement of traffic rules. For example, by subjecting motorists to extensive auto school training including information on cycling safety.

Source: The PEP Partnership on Cycling (2021[32]), "Toolbox for Action for Cycling Promotion based on best available experience from the country of the Pan-European region: Annex 1 of the Pan-European master plan for cycling promotion, May 2021", https://thepep.unece.org/sites/default/files/2021-06/TOOLBOX_2021-05-20-II_BF%203%20June_0.pdf.

Parking policies

Parking policies can either incentivise or disincentive the use of cars in urban areas. Parking policies that yield low- or under-pricing of parking facilities increase demand for car use at the expense of other health and environmentally friendly transport modes. Parking facilities also incur a high social cost – for example, due to biodiversity losses as well as loss of open space that could alternatively be used to create green, recreational areas (The PEP Partnership on Cycling, 2021[32]).

To reduce car use, parking tariffs should reflect both their economic and social cost, which for many cities would increase the cost of parking. Other parking policies to reduce car use include (Russo, Van Ommeren and Dimitropoulos, 2019[33]):

- Smart parking meters, which allow tariffs to change according to the location, and time of the day/week. The purpose of these meters is to keep parking occupancy rates below 100% so that drivers can always find a park and therefore spend less time "cruising for parking".
- Stricter enforcement through frequent monitoring of parking space use in addition to higher fines for violations.
- Adopting maximum parking requirements – i.e. the maximum number of parking spots for a given area.
- Offering employees parking "cash-outs" that is equivalent to the cost of subsidising workplace parking.
- Zoning regulations (e.g. only allow lightweight vehicles in parts of the city during certain times)
- Removing selected parking spots and limiting the number of registered spots per person (see Box 1.5 for an example from Paris).

Box 1.5. Paris parking reforms

In urban area such as Paris, car parking space still accounts for 50% of public space, despite car use making up on 10% of all trips. In response, policy makers in the French capital aim to transform half of all existing parking spots (which is equivalent to 60 hectares) into sidewalks, cycle paths, green spaces etc. Further, remaining parking spots will be prioritised for members of the public with priority needs such as people living with a handicap as well as certain professions (e.g. health professional's home visits).

Additional planned changes as part of the reform include:

- Free parking for those with reduced mobility
- Free parking for low-emission vehicles
- Free parking for certain professionals (e.g. nursing staff).

Source: Ville de Paris (2022[34]), "Paris reforming its parking lot", https://www.paris.fr/pages/paris-reforme-son-stationnement-17981.

Road pricing

Road pricing schemes impose a fee for drivers to use roads at certain time of day, and are dependent on the area. For example, in London, drivers must pay GBP 15 a day if they enter a Congestion Charge zone between 7:00-18:00 Monday to Friday, and between 12:00-18:00 Saturday to Sunday. Congestion charges

aim to manage existing roads and contribute to decongestion issues within a city. According to the International Transport Forum, well-organised and executed road pricing schemes reduce the need to expand road capacity (International Transport Forum, 2018[35]), thereby leaving more space for pedestrians, cyclists, green space etc.

It is important to note that since 2020 Barcelona has a road scheme in place that aims to reduce carbon emissions – specifically, cars without a Directorate General for Traffic environmental label cannot drive in the city between Monday and Friday, 7:00-18:00.

Other schemes to either disincentive car use or incentivise active modes of transport

Policies related to cycling, parking and road pricing represent just a few of the many ways to reduce air pollution by promoting active modes of transport. Other measures at the disposable of policy makers, which are not discussed in detail are listed below:

- Establish bike parking facilities at major train stations, for example, to promote multi-modal travel – for example, the Utrecht Central Station in the Netherlands includes room for 12 500 bikes.
- Encourage carpooling (e.g. through separate traffic lanes to reduce commuting time)
- Flat rate distance-based charges for car drivers
- Banning highly-polluting vehicles from the road
- Tax-deductible public transport costs.

Enhancing equity

Initially, implementation of Superblocks areas occurred on one neighbourhood at a time basis. However, this had a negative effect on neighbouring areas who felt discriminated against – e.g. due to more traffic and noise in their area, as well as difficulty travelling to areas on either side of the new superblock. To limit inequalities between neighbourhoods, Superblock administrators abandoned their incremental implementation approach and replaced it with a vision to create a “network of green streets and squares” (Staricco and Brovarone, 2022[36]). The Eixample area in Barcelona tested this new implementation approach, which is now being scaled-up across the whole city.

Policy makers should continue efforts to minimise negative externalities felt by neighbouring Superblock areas. Failing to do so will led to unintentional inequalities across Barcelona. Further, it may stall implementation efforts if the public are dissatisfied.

Enhancing the evidence-base

Evaluating the impact of urban design projects on physical activity is difficult for various reasons – e.g. large populations covered, multitude of confounding factors, and difficulties obtaining objective outcome measures. This study evaluated the impact of Superblocks using a modelling exercise based on assumptions of how Superblocks affected time spent being active. Although useful, modelling exercises only provide an indication of the future benefit of Superblocks as opposed to the intervention’s real impact. Future evaluations based on real-world data would therefore enhance the evidence-base supporting Superblocks.

The remainder of this section outlines information and tips for undertaking a robust evaluation of urban design projects, such as Superblocks, using real-world data. It covers the study design; indicators and data sources; collaboration across research disciplines; as well as additional resources for further information.

Study design

Evaluations of similar urban design interventions, such as Complete Streets and Move the Neighbourhood, conducted in the United States and Denmark, respectively, used quasi-experimental study designs to measure their impact on physical activity (Brown et al., 2016[37]; Pawlowski et al., 2017[38]). Quasi-experimental studies, although not as robust as randomised-control trials (RCTs), have many benefits, such as those outlined below:

- Avoid ethical concerns, which can arise from randomly allocated sections of the population into treatment and control groups
- Less expensive and require fewer resources than RCTs
- Able to measure the real-world impact of interventions, therefore results are more generalisable
- Allow researchers to measure the change in key outcome measures over time between two groups (i.e. those that access the intervention and those who do not).

Indicators and data sources

At present, it is difficult to compare interventions that promote active transport both within and across countries due the absence of comparable data. This is problematic as it prevents researchers from understanding the relative effectiveness of different interventions, as well the impact of interventions on different population groups.

Future evaluations of Superblocks should consider measuring a change in physical activity using existing indicators such as those collected by Euro-barometer, the European Health Interview Survey (EHIS) as well as universally recognised indicators from the WHO (e.g. time per kilometre spent walking and/or cycling per person per day for all travel purposes) (WHO Europe, 2021[39]).

Data for these indicators, to the extent possible, should be objective as they are more reliable and valid. There are various ways to collective objective physical activity measures predominantly from either a wearable pedometer or accelerometer (activity watch). For example, the evaluation Denmark's "Move the Neighbourhood" intervention used data collected from accelerometers and GPS to assess where participants spent their time and how active they were (Pawlowski et al., 2017[38]).

Subjective measures however may be necessary and offer many benefits – e.g. they are practical, versatile, relatively inexpensive and easy to use. Subjective measures are typically collected using self-reported physical activity questionnaires and activity diaries/logs.

Collaboration across research disciplines

Information on study design, indicators and data sources focuses on evaluation from a public health point of view. However, there are various ways to measure the impact of changes to the built environment on obesity. For example, other disciplines interested in this topic include urban planners, transport, epidemiology, and behaviour and social sciences. These disciplines evaluate changes to the built environment from a different perspective using different measurement tools (see Table 1.4). Future evaluations, which includes experts from these disciplines, may better capture the true holistic impact of Superblocks.

Table 1.4. Types of disciplines involved in evaluating built environment changes

Discipline	Importance or emphasis	Examples of key concepts
Urban Planning	Focuses on the technical and social-political processes that shape land-use patterns and community design.	Use of geographic information systems (GIS) to extract measures of density and land-use mix from existing

Discipline	Importance or emphasis	Examples of key concepts
		data sources. Measures such as walkability draw on urban planning concepts
Transportation	Planning and design of physical infrastructure of roads, sidewalks, bike paths, railroad tracks, bridges, etc.; understanding of daily travel choices	Use of GIS to extract measures of features of transportation systems. Measures of connectivity of street and pedestrian/bike systems.
Nutritional Science, Public Health Nutrition	Focus on how food choices are affected by neighbourhood, store/ restaurant and home food environments	Local/setting availability may influence what people eat. Measures: NEMS-S, NEMS-R
Exercise Science and Physical Activity Research	Attention to physical activity environments in neighbourhoods and organizational settings where PA occurs	Self-report surveys, systematic observations, and secondary data analysis of walkability, bikeability, off-road walking/biking trails, parks and other physical activity settings. Measures: MAPS, SOPARC, SOPLAY
Epidemiology and Public Health	Study of the patterns, causes, and effects of health and disease conditions. Informs policy decisions and evidence-based practice. Often emphasizes health disparities	Contributes to study design, collection and statistical analysis of data, and interpretation and dissemination of results. May involve linking population-based behavioural or biological data to environment assessments.
Behavioural and Social Sciences	Examines behaviour and processes and social context, societal level variables, and relationships within community. Emphasizes the importance of place to health.	Strong expertise in assessment, measurement development and psychometrics, experimental design, multi-level analysis and complex modelling. Measures: Wellness Child Care Assessment Tool, NEMS-P
Economics	Examines the importance of environmental factors including pricing, taxation and marketing on food consumption, PA behaviours and health-related outcomes. Often emphasizes health disparities	Linkage of individual-level and contextual data sets with emphasis on fixed effects and longitudinal models. Focus on how enactment of policies affect the BE through changes such as food pricing, availability, and advertising. Tools/measures: Price elasticity of demand, BTG-COMP, secondary data analysis.

Source: Text taken from Glanz et al. (2016[40]), "Built environment assessment: Multidisciplinary perspectives", https://doi.org/10.1016/j.ssmph.2016.02.002.

Additional resources

Listed below are additional documents for further information on how to evaluate changes to the built environment:

- OECD's Guidebook on Best Practices in Public Health (OECD, 2022[41])
- WHO's health and economic assessment tool (HEAT) dedicated to assessing the economic benefits arising from cycling and walking (World Health Organization, 2017[42]).

Enhancing extent of coverage

This section outlines several ways to assist the expansion of Superblocks across Barcelona: involving stakeholders at an early stage, promoting evaluation findings, as well as targeting people from a young age. These same principles apply to all forms of urban regeneration projects.

Local stakeholder involvement from an early stage

The Superblocks approach alters the built environment, which affects the day-to-day lives of residents, businesses, local governments, and many other groups. Therefore, it is essential these stakeholder groups be involved in the decision-making process from the beginning, and are informed on the project's objectives

and how these address community needs. Including stakeholders during early stages, however, is not sufficient; it is also essential that stakeholders understand the changes proposed under the project and that their opinions are heard and acted upon. By promoting active stakeholder participation in the decision-making process, Superblocks is more likely to receive "buy-in" from the community, as well as minimise implementation barriers and maximise outcomes. For example, involving (Shimomura and Matsumoto, 2010[43]):

- Local residents helps ensure needs are met at the local level, thereby preventing issues associated with gentrification and displacement
- Local government representatives helps ensure timely identification of local points of tensions as well as solutions.

For stakeholder participation to be effective, stakeholders must receive all necessary information on the planned changes. Highlighting the importance of this information is also necessary so that it is not disregarded (Shimomura and Matsumoto, 2010[43]).

Promoting evaluation findings

Implementation of large urban regeneration projects often occurs in stepped manner – i.e. area by area. This provides an excellent opportunity to evaluate pilot areas to assess if project objectives are being met, such as a reduction in pollution, higher life satisfaction and more active modes of transport. Positive evaluation findings are useful to justify continued funding for the project as well as encourage reluctant stakeholders to buy in to the project (OECD, 2022[41]). There is no guarantee evaluations will result in positive findings, even if the project is designed based on best practice principles. They are nonetheless important as they provide crucial information on how to design, deliver and improve a project, which also leads to an increase in stakeholder support.

Target residents from a young age

During younger adulthood (16-35 years), people experiment with different travel options until they find one that positively fits with their lifestyle. This mode of transport is then typically used throughout the person's working life (Pistoll and Cummins, 2019[44]). By this reasoning, policies to promote active modes of transport under Superblocks should target older school-aged children in order to cultivate life-long healthy behaviours (e.g. cycling and/or walking to work).

1.3. Transferability

This section explores the transferability of Superblocks and is broken into three components: 1) an examination of previous transfers; 2) a transferability assessment using publically available data; and 3) additional considerations for policy makers interested in transferring Superblocks.

Previous transfers

The following section outlines the expansion of Superblock areas across Barcelona, as well as similar urban regeneration projects implemented in other OECD countries.

Within country transfers

The city of Vitoria-Gasteiz implemented the first superblock in 2008. An evaluation of this pilot project found superblocks reduced greenhouse gas emissions and noise levels while simultaneously increasing the number of people switching from using their car to either cycle of walk. The evaluation also outlined a list of "lessons learnt" for future transfer projects, such as (Albaina, 2019[45]):

- Ensuring plans for the city align are based on the same foundations as those for superblocks – i.e. major infrastructure change, powerful participation, communication and promotion campaigns
- Taking any opportunity to transform urban space to accommodate the superblocks design, even before implementation (e.g. during necessary roadworks).

A further five superblocks have been implemented since the first pilot in 2008. To assist with the transfer process across different neighbourhoods, city administrators conduct necessary measures in the neighbourhood prior to implementation – for example (Staricco and Brovarone, 2022[36]):

- Engaging local residents and businesses in the decision to participate
- Canvassing the needs of a specific area and of the local community.

Box 1.6. Let's fill streets with life

The "Let's fill streets with life" is a pre-implementation program that was designed as a primary step in moving to actionable measures in establishing the Superblocks model in the city of Barcelona. The purpose of this program is to carry out a diagnosis of the area that has been earmarked for urban regeneration efforts through technical evaluation, public participatory meetings and road mapping for the implementation of appropriate initiatives. The Commission for Ecology, Urban Planning and Mobility has proposed a methodology to achieve this preliminary preparation process, by answering four targeted questions:

Where is it to be established?

- Territorial areas of action are areas of various sizes that include zones with formal or functional features that are permit for Superblocks to initiate in relation to the basic streets that they cross.

Who will be accompanying the development?

- Involvement of key territorial players such as local residents, both city and private associations, districts, relevant agencies and experts.

What will be the nature of the initiatives?

- A city is not a homogenous entity and carries different morphological and social considerations, which requires for urban regeneration initiatives to have various levels of action, for instance functional, tactical and structural level changes.

What will be the initiative-defining process be like?

- The process for defining initiatives appropriate for implementation of Superblocks model in public spaces consists of five successive work stages, divided into two phases: examination phase (stage 1 and 2) and implementation phases (stage 3 to 5). Stage 1: examines and applies technical criteria; Stage 2: defines the Action Plan for the specific superblock area; Stage 3: prioritises the initiatives proposes under the Action Plan; Stage 4: Defines the executive projects for the Action Plan's initiatives; Stage 5: implementing the initiatives.

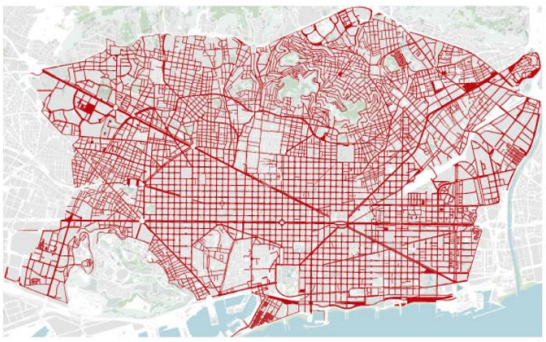
Source: Ajuntament de Barcelona (2016[55]) https://ajuntament.barcelona.cat/ecologiaurbana/en/bodies-involved/citizen-participation/superblocks

International transfers

.~~The Superblocks approach exists only in Spain. Due to the unique layout of Barcelona (i.e. grid like layout), transferring this intervention to other major cities across the world requires careful consideration.~~

~~Significant adjustments are likely to be necessary. In fact, transferring any form of urban regeneration project requires major research given cities vary in their scale, population density, urban shape and form, development patterns and institutional frameworks (Nanda, 2019[46])~~. IT IS NOT correct!!!. Several cities in several countries are developing the superblock concept. I propose this new text:

The superblocks model urbanizes the car (the time to go around a superblock by car is similar to the time needed to go around a block on foot). It is a model that can be perfectly transferable to any city in the world, regardless of its morphological characteristics, densities, etc. The superblock model consists of a network of roads that defines the perimeter of the superblocks and integrates the set of transport modes. Cars, public transport, the main bicycle network, and pedestrians circulate along the perimeter roads that define the superblocks and act as connecting pathways. The aim of this network is to free up the maximum number of streets currently dedicated to passing mobility for other uses (70-75% of the road surface is freed from traffic and will be pedestrian priority), and, at the same time, guarantee the functionality of the city by reducing a minimum percentage of traffic, between 10 and 15%, depending on the morphology of the city. With this reduction in traffic, the superblock model achieves a traffic service, that is, an average speed in the city, similar to the average speed in the current scenario. In Barcelona (BCN) a network of 503 superblocks would free up 6,200,000 m2 of road surface (70% of the road space will be pedestrianized) with a 15% reduction in traffic. In Vitoria, 75% of the road would be liberated with a 12% reduction in traffic. Rueda, S, (2020).

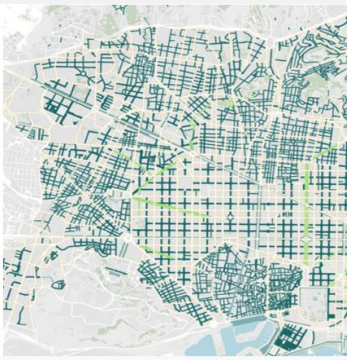


BCN streets for road traffic: current scenario

Source: Rueda, S, (2020)

BCN streets for road traffic: superblock scenario

Source: Rueda, S, (2020)



BCN streets with pedestrian priori

Source: Rueda, S, (2020)

Each city, depending on its morphology and the characteristics of the main roads will have its own network, that is, the superblock model is perfectly transferable to any city in the world.

Although the model is transferable to any city, ~~Given each city is unique,~~ it is not surprising that OECD countries have taken their own urban regeneration approach – see Box 1.7 for examples in Paris, London and Hamburg. Although approaches differ across cities, they maintain the same core objectives such as reducing air pollution, prioritising pedestrians, increasing green spaces, and improving population wellbeing.

Box 1.7. Similar urban regeneration schemes in Europe

Paris, France

15 minutes city

Paris recently introduced the concept of a "15-minute city", whereby work, schools, shops, entertainment, culture, leisure and other activities are accessible within a 15 minute walk or a five minute bicycle ride (Ancona, 2021[47]). The aim is to decentralise services within the city, and subsequently reduce car circulation resulting in a more active, green, flexible and heathy city (Ancona, 2021[47]).

Health impact evaluations for this initiative are yet to be undertaken, but projections indicate the concept will decrease air and noise pollution as well as CO_2 emissions.

London, United Kingdom

Low-traffic neighbourhoods of London (London Living Streets)

To encourage Londoners to spend more time on waking, and provide safer environment for residents (both walkers and cyclists), some simple circulation changes were implemented in urban streets (Aldred and Goodman, 2020[48]). The aim was to:

- Limit traffic in smaller residential streets
- Provide more opportunities for physical activity outdoors
- Reduce crime
- Encourage socialising.

Early impact evaluations found affected residents engaged in 41 additional minutes of walking or cycling per week (Aldred, Croft and Goodman, 2019[49]).

Hamburg and Freiberg, Germany

Car free cities and neighbourhoods

The German cities of Hamburg and Freiberg are adopting car free city models, which aim to reduce congestion and improve the quality of life for its citizens. This involves making streets walking and cycling centric, while investing in neighbourhoods without cars and sustainable housing.

This new city model aims to:

- Allow residents across all SES areas be able to walk to grocery stores, cafes and restaurants, workplaces, and schools for all levels of SES in the area (Peters, 2019[50]).
- Increase the proportion of urban area covered by parks from 16.8% to 40% by 2035, while simultaneously reducing CO_2 footprint by 80% (Glazener and Khreis, 2019[51]).

Transferability assessment

Methodological framework

Details on the methodological framework to assess transferability are in Table 1.5. Indicators from publically available datasets to assess the transferability of Superblocks are listed in. These cover indicators related to the population, political and economic contexts. Please note, the assessment is

intentionally high level given the availability of public data covering OECD and non-OECD European countries.

Table 1.5. Indicators to assess the transferability of Superblocks

Indicator	Reasoning	Interpretation
Population context		
% of people living in an urban area	A metric that measures the proportion of the population that is eligible to benefit from improved mobility infrastructures in urban areas	↑ value = more transferable
Overall City Satisfaction	Superblock is more likely to be well transferable to city/country where people are less satisfied with their city as it makes them more open to major changes introduced to the city	↓ value = more transferable
Satisfaction of Public Transportation Systems	Superblock is more likely to be well transferred to city/country where people are less satisfied with the transportation system currently in place as it makes them more open to major changes introduced to the mobility infrastructure	↓ value = more transferable
Sector context (urban projects)		
Active transport national scheme	Superblock is more transferable in countries that have schemes in place that prioritise active transport to work for the population	'Implemented' = more transferable
Transport type	Superblocks is more transferable countries with existing culture of use of non-motorized (bicycles) or smaller motorized (mopeds, scooters) mode of transport	↑ value = more transferable
Transport friendly attitude	Superblock is in part behaviour change intervention, where the positive attitudes of the population in switching their urban mobility patterns will permit for stronger reception of urban environmental changes that promote use of public and active transports	↑ value = more transferable
Political context		
Environmental policy stringency index	Superblock will be more transferable to countries that rank in the highest degree of stringency for national policies with implicit or explicit price on pollution or environmentally harmful behaviour	↑ value = more transferable
National policy on air pollution reduction	Superblocks aims at reducing various levels of pollution (air, heat and sound), hence it is more transferable to countries that have commitments through legally binding standards and regulations.	'Implemented' = more transferable
Economic context		
GNI per capita	As Superblocks includes investment in various spheres of urban organisation, countries that have more resources available for transport and are more likely to have other pressing societal issues addressed	↑ value = more transferable

Source: World Bank (2018[53]), Gallup Analytics (2021[54]), WHO (2018[55]), Eurobarometer (2019[56]), OECD (2015[57]), Transport Policy (2018[58]), World Bank (2017[59]).

Results

Results from the transferability assessment using publically available data are summarised below, which show mixed results (see Table 1.6 for results at the country level):

- Spain's urban population is higher than the average OECD/EU country (81% vs 78%), indicating certain countries may find it difficult to implement large-scale urban regeneration projects, such as Superblocks.
- The vast majority of countries had the same or lower levels of satisfaction with their city or public transport system when compared to Spain. This may mean they would be more open and supportive of urban environment changes, thus more accepting of the Superblock model.
- All OECD and non-OECD European countries with available data have air pollution regulations, standards or policy in place that would provide political support to the Superblock model.

- A large proportion of citizens (70%) in Spain report being motivated to make the switch from private vehicle to active modes of transportation, compared to an average of 58% among potential transfer countries. This is an indicator for some countries that there may be difficulty encouraging people to embrace Superblocks.
- Spain does not have a targeted national level scheme for promoting active transport to work, compared to 15 analysed countries who either have a scheme or foresee one in the future. An active transport scheme is important for obtaining support urban regeneration efforts.

Table 1.6. Transferability assessment by country (OECD and non-OECD European countries)

A darker shade indicates Superblocks model is more suitable for transferral in that particular country

Country	People living in urban areas (%)	City Satisfaction (%)	Public Transport Satisfaction (%)	National schemes for active travel to work	Respondents who take active transport (%)	Population ready to switch to active transport (%)	Environmental stringency index	Air pollution policy	GNI per capita, PPP
Spain	**81**	**86**	**68**	**Not Implemented**	**3**	**70**	**2.22**	**Implemented**	**41046.44**
Australia	86	91	63	n/a	n/a	n/a	3.17	Implemented	48007.15
Austria	59	92	74	Implemented	9	54	2.95	Implemented	56304.49
Belgium	98	89	63	Implemented	13	56	2.47	Implemented	52561.54
Bulgaria	76	n/a	n/a	Foreseen	3	41	n/a	n/a	22883.44
Canada	82	87	57	n/a	n/a	n/a	3.28	Implemented	48383.9
Chile	88	82	62	n/a	n/a	n/a	n/a	Implemented	24130.56
Colombia	81	80	59	n/a	n/a	n/a	n/a	Implemented	14163.25
Costa Rica	81	86	69	n/a	n/a	n/a	n/a	Implemented	19094.39
Croatia	58	77	48	Not Implemented	8	61	n/a	Implemented	28388.17
Cyprus	67	92	48	Not Implemented	n/a	61	n/a	Implemented	38206.84
Czech Republic	74	91	77	Implemented	7	50	2.38	Implemented	38326.49
Denmark	88	96	73	Implemented	24	59	3.85	Implemented	57449.41
Estonia	69	92	62	Not Implemented	5	60	n/a	Implemented	36122.94
Finland	86	91	63	Implemented	13	61	3.43	Implemented	49050.05
France	81	84	63	Implemented	4	62	3.58	Implemented	47065.15
Germany	77	90	64	Implemented	15	59	3.13	Implemented	54877.89
Greece	80	77	43	n/a	3	65	2.13	Implemented	29707.77
Hungary	72	79	61	Implemented	16	42	2.63	Implemented	31770.52
Iceland	94	82	59	n/a	n/a	n/a	n/a	Implemented	54095.41
Ireland	64	88	61	Implemented	1	59	2.05	Implemented	65697.68
Israel	93	83	62	n/a	n/a	n/a	n/a	Implemented	39945.9
Italy	71	71	41	Not Implemented	0.06	59	3.28	Implemented	42784.39
Japan	92	85	62	n/a	n/a	n/a	3.17	Implemented	42808.04
Latvia	68	89	59	Not Implemented	0.09	58	n/a	Implemented	30527.64
Lithuania	68	88	47	Not Implemented	0.06	41	n/a	Implemented	35989.23
Luxembourg	91	95	79	Implemented	0.03	77	n/a	Implemented	72375.57
Malta	95	n/a	n/a	Not Implemented	2	70	n/a	n/a	40371.94
Mexico	81	77	57	n/a	n/a	n/a	n/a	Implemented	19188.55
Netherlands	92	95	78	Implemented	41	68	3.63	Implemented	57071.77

New Zealand	87	90	56	n/a	n/a	n/a	n/a	Implemented	41671.54
Norway	83	96	68	n/a	n/a	n/a	3.26	Implemented	67562.82
Poland	60	88	60	Not Implemented	10	50	2.58	Implemented	31913.42
Portugal	66	94	58	Not Implemented	n/a	39	2.13	Implemented	34154.42
Republic of Korea	81	0.82	75	n/a	n/a	n/a	3.07	Implemented	43240.25
Romania	54	n/a	n/a	Not Implemented	7	63	n/a	n/a	29549.19
Slovak Republic	54	85	56	Implemented	7	56	2.99	Implemented	29622.32
Slovenia	55	92	60	Implemented	6	65	2.28	Implemented	38411.08
Sweden	88	95	70	Not Implemented	21	71	3.1	Implemented	53928.41
Switzerland	74	93	87	n/a	n/a	n/a	3.29	Implemented	65821.31
Turkey	76	65	52	n/a	n/a	n/a	1.92	Implemented	27813.76
United Kingdom	84	83	68	Implemented	3	56	3.83	Implemented	45850.64
United States	83	80	60	n/a	n/a	n/a	2.69	Implemented	62512.91

Note: *n/a = not available data. The shades of blue represent the distance each country is from the country in which the intervention currently operates, with a darker shade indicating greater transfer potential based on that particular indicator (see X for further methodological details).
Source: World Bank (2018[53]), Gallup Analytics (2021[54]), WHO (2018[55]), Eurobarometer (2019[56]), OECD (2015[57]), Transport Policy (2018[58]), World Bank (2017[59]).

To help consolidate findings from the transferability assessment above, countries have been clustered into one of three groups based on indicators reported in Table 1.6. Countries in clusters with more positive values have the greatest transfer potential. For further details on the methodological approach used, please refer to X.

Key findings from each of the clusters are below with further details in Figure 1.10 and Table 1.7:

- Countries in cluster one have population, sector specific (i.e. urban environment), political and economic arrangements in place that would support the Superblock model. These countries, therefore, are less likely to experience implementation barriers should they choose to transfer Superblocks or something similar.
- Countries in cluster two also have sector population, political and economic arrangements in place that support the transfer of Superblocks but could consider further action to ensure the population are ready to switch to more active modes of transport. It is important to note that Spain falls under this cluster meaning conditions in which these countries could improve on, although ideal, are not pre-requisites.
- Countries in cluster three should consider a number of factors before transferring Superblocks model such as ensuring Superblocks aligns with overall political priorities and is affordable (i.e. there is sufficient funding to prioritise funding for transport changes).

Figure 1.10. Transferability assessment using clustering

a World Bank (2018[53]), Gallup Analytics (2021[54]), WHO (2018[55]), Eurobarometer (2019[56]), OECD (2015[57]), Transport Policy (2018[58]), World Bank (2017[59]).

Table 1.7. Countries by cluster

Cluster 1	Cluster 2	Cluster 3
Austria	Australia	Bulgaria
Denmark	Belgium	Chile
Germany	Canada	Colombia
Ireland	Cyprus	Costa Rica
Luxembourg	Czech Republic	Croatia
Netherlands	Estonia	Greece
Norway	Finland	Hungary
Sweden	France	Mexico
Switzerland	Iceland	Romania
United States	Israel	Slovak Republic
	Italy	Turkey
	Japan	
	Latvia	
	Lithuania	
	Malta	
	New Zealand	
	Poland	
	Portugal	
	Republic of Korea	
	Slovenia	
	Spain	
	United Kingdom	

Note: http://oe.cd/cyprus-disclaimer & http://oe.cd/israel-disclaimer

New indicators to assess transferability

Data from publically available datasets is not ideal to assess the transferability of Superblocks, for example, there is no publically available data across on current public transport infrastructure. Therefore, Box 1.8 outlines several new indicators policy makers should consider before transferring this intervention.

Box 1.8. New indicators to assess transferability

In addition to the publically available data within the transferability assessment, policy makers are encouraged to collect data for the following indicators that have been outlined within the European Commission's "Physical activity through sustainable transport approaches: case study cities" indicators (PASTA, 2014[60])

Population context

- How satisfied are the urban population with public park?
- What is the turnover rate of overnight tourism in the city?
- What proportion of urban population lives 500m from public transport stop?

Sector-specific context (urban projects)

- What infrastructure is available in the city to encourage adopting active transport modes?
- What percentage of jobs and urban services are accessible within 60-minute journey by public transport?
- How well does the geographical level data collection permits to visualise regional level detail?

Political context

- How well does the city involve citizens and stakeholders in the decision-making and delivery of active mobility?
- Has the intervention received commitment from key decision-makers?

Economic context

- Is there funding for environmental protection and transport mobility projects? (E.g. European Commission Cohesion Fund)

Source: European Commission (2020[54]) https://ec.europa.eu/regional_policy/en/funding/cohesion-fund/, PASTA (2014[60]) http://www.ecomobility-shift.org/en/, Polivisu (2019[55]) https://policyvisuals.eu/visualisation-type-chloropleth-map/

1.4 Conclusions and next steps

Superblocks model is an urban regeneration intervention targeting cities with densely populated urban areas. The purpose this urban environment modification model is the reduction of air pollution, noise pollution and increasing use of active transport modes in densely populated urban areas. The Superblocks model was piloted in city of Vitoria-Gasteiz in Northern part of Spain, and was adapted to larger and denser city of Barcelona.

Estimates indicate scaling up of Superblocks across all of Spain would lead to significant health and economic gains. Superblocks would yield health expenditure saving of EUR 4.04 per person, or EUR 0.26 ~~EUR~~ per person per year. Additionally, this intervention was deemed as cost-effective under the modelling work across OECD and non-EU countries.

Superblocks has the potential to narrow health inequalities, especially among population with low SES as well as increase social cohesion between generations. Superblocks model focuses on improvement of city's pollution and noise levels by reducing their intensity, populations with low SES often live in areas with heightened levels of both air and noise pollution. Additionally, through structural and

functional changes of the neighbourhoods, the aim is at promoting further social cohesion via provision of safe and accessible shared space to people of all ages.

The number of well-established and functional Superblocks is continuously growing in city of Barcelona. Between the inceptions of Superblocks in 2014 to today, there are six fully active Superblocks, with over 500 being planned to take up shape throughout the expansion efforts in the city of Barcelona.

Superblocks model has a positive impact on many best practice criteria; however, further enhancements are possible. For instance, policymakers could promote complementary policies and actions through changes to parking regulations and road pricing policies, additionally inclusion of cycling in road safety policies aimed both children and adults population. Furthermore, in order to evaluate accurately the positive effects on health status from environmental changes through use of targeted indicators, adapted study designs and use of advanced tools (accelerometers and mobility surveys) for reliable data.

Urban environment regeneration interventions similar to Superblocks model exists in many OECD countries, indicating the transferability of this intervention. Although the exact model of Superblocks ~~for the moment is unique to Spain~~ IT IS NOT CORRECT, other European cities such as Paris, London and Hamburg, are engaged in promotion of similar measures whose ultimate goal is the reduction of air and noise pollution, increase in road safety for higher uptake of active and public transport and overall city satisfaction. Transferability of Superblocks was also accessed using publically available data, which found mixed results – for example, Superblocks would likely receive political support in most countries given that OECD countries have climate change as a top priority, nevertheless not all countries have a strong regulatory schemes for active travel to work.

Box 1.9. Next steps for policy makers and funding agencies

Next steps for policy makers and funding agencies to enhance Superblocks model are listed below:

- Support policy efforts to establish and uphold road safety rules for all types of vehicle users (motorised or non-motorised modes of transport), by example including comprehensive education and practice to road schools for both drivers and pedestrians.
- Support policies that aim at making use of gas emitting modes of transport more expensive and inconvenient to serve as deterrent and lead to adoption of active modes of transport.
- Ensure establishment of nationwide active transport schemes or policies that promotes parity through uniformity of measures throughout the country

Enhance support for policies and strategies that aim at improvement of lived environment, especially in low-SES areas (e.g. green spaces, equitable access to healthy food options)

Promote ‘lessons learnt’ from regions that have transferred Superblocks model to their local setting.

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Annex A. Modelling assumptions

The SPHeP-NCD model relied on several inputs and assumptions as outlined in the table below.

Table A A.1. Parameters to model the impact of Superblocks

Model parameters	Superblocks model inputs
Effectiveness	Average increase in public transport related MET minutes per week of 894.98. Based on the methodology by (Graf and Cecchini, 2019[56]), this figures was translated into change in total physical activity X (given there is evidence suggesting an increase in transport related physical activity results in a decrease in other forms of physical activity). See subsequent tables for further information.
Target age	Men and women aged 20 years and over based on (Mueller et al., 2020[29]).
Coverage;:/Exposure	Based on (Mueller et al., 2020[29]) 19.2% of the population will switch to active modes of transport as a result of Superblocks. Not everyone lives in an area that is suitable for Superblocks, therefore, the eligible population is restricted to those who live in urban areas (data on % population in urban areas was taken from World Bank – "Urban population (% total population)". It is unlikely that all urban areas will be suited for Superblocks, for this reason, we conservatively estimate that only 50% of all urban areas in a country are eligible. In case of Spain, with the coverage of 8% of population Target for coverage is country specific. In addition, we take into account that it takes many years to implement an intervention like Superblocks. For this reason, we use a conservative estimate of 15 years – i.e. coverage will increase linearly from 0% to 100% over a 15-year period.
Per capita and participant cost, EUR	Based on a (Silverton and Spath, 2021[64]) report which states that cost of one Superblock, an estimate to be EUR 100 million for all of the 503 planned superblocks, therefore, the be propose an assumption that this costs will distributed over a 15-year period. Hence the cost per capita per annum for the population (1.62 million) of Barcelona = EUR 4.12 per capita of per annum for all of city of Barcelona Scenarios of proportion of costs being attributed to health outcomes (per capita) – an assumption: 15% of costs attributable to health outcomes: EUR 0.62 per capita 25% of costs attributable to health outcomes: EUR 1.03 per capita 50% of costs attributable to health outcomes: EUR 2.06 per capita

The following table reflects data from (Mueller et al., 2020[29]) outlining how people's transport preferences change due to Superblocks and the related change in MET minutes.

Table A A.2. Average MET minutes per week by transport type

Superblocks MET gains (a)	% population that switch from motorised vehicles to active transport* (b)	Mean duration of activities gained (c)	Trips per day (d)	Days per week (e)	MET-minutes/week increase (= a * c * d * e)

Public transport = 3.5 METs	7% ((= 23,433 / 65,092)*0.19)	10 minutes	3.5	5	= 612.5
Bicycle = 6.8 METs	2% ((=5,208 / 65,092)*0.19)	16.8 minutes	3.5	5	= 1999.2
Walking = 3.5 METs	11% ((=36,451 / 65,092)*0.19)	15 minutes	3.5	5	= 918.8

Note: *Based on (Mueller et al., 2020[29]), 19.2% (= 65,092 people) of the population will switch to either public transport, a bike or walk as a result of Superblocks.

The three different MET minutes per week figures were transformed into one weighted average MET increase due to changes in transport related physical activity.

Table A A.3. Weighted average MET minutes per week

Transport type	% of all people who switch according to type of transport (a)	Weekly increase in MET minutes (b)	Weighted average (a) * (b)
Public transport	36% (=7%/19.2%)	612.5	220.50
Bicycle	8% (=2%/19.2%)	1999.2	159.96
Walking	56% (=11%/19.2%)	918.8	514.52
Total			**220.5 + 159.96 + 514.52 = 894.98**